

MESSI vs RONALDO

Statistics Explained

A plain-English teaching guide through verified match-level uncertainty

THE QUESTION What changes when goals are compared per 90 minutes and then adjusted by a statistically derived match-context index?

What this guide covers

The collected sources, the cleaning path, every completed formula and metric, the current results, and the limits of the evidence through Dashboard Wave 6.

NEUTRAL • DATA-SNAPSHOT RESULTS • NOT A WINNER DECLARATION

Snapshot	Value
Goals	1,738
Valid appearances	2,201
Valid minutes	181,081
Coverage status	Wave 6 complete and verified

Contents and the 60-second explanation

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The 60-second explanation

We collected 1,738 goals and 2,201 valid appearances for Messi and Ronaldo. Every valid minute, including 1,063 scoreless appearances, stays in the per-90 denominator. One global FAMd combines opponent Elo, venue, competition stage, and an away indicator into a centred context index. The K control applies signed power. The Inference tab freezes the clicked K, penalty, competition, and venue settings, then independently resamples whole appearances within each player 10,000 times. It reports the full Messi-minus-Ronaldo gap distribution, a percentile interval, directional probability, and a conventional match-level Cohen's d without p-values or a winner flag.

SAY IT SIMPLY We compare scoring rates using all playing time, then stress-test a data-derived context index without pretending the index is a literal goal count.

How to read the numbers

- Raw goals per 90 are literal scoring rates.
- Weighted values are centred index units per 90, so they can be negative.
- The comparison is a fixed dataset snapshot, not live official career totals.

What was collected-and what currently affects the statistics

Source	Collected	Why collected	Current statistical role
FBref goal logs	1,738 goal rows	One row per goal event	Yes-goal numerators and goal context
FBref match logs	2,263 raw rows	Appearances, minutes, scoreless matches	Yes-FAMD and all per-90 denominators
FBref season stats	140 rows	Season summaries and source checks	No direct role in current weighting
Understat	487 xG goal records; 483 matched	Shot expected-goal context	Attached, but not used in difficulty weight
ClubElo	633,265 lookup rows	Historical club opponent ratings	Yes-first Elo source for club matches
National Elo	1,820 lookup rows	Historical national-team ratings	Yes-for international matches
Transfermarkt transfers	20 rows	Club-movement history	No effect on current weighting
Transfermarkt biography	0 usable rows	Biographical details	No effect on current weighting

Coverage that must not be confused

- Understat supplied xG for 483 of 1,738 goals; 1,255 goals lack xG.
- xG is descriptive enrichment only. The dashboard difficulty score does not use xG.
- Transfer rows can explain career journeys but do not enter FAMD, raw rates, weighted rates, or the K test.

SAY IT SIMPLY Collected does not mean used. The current weight uses match context-not xG, transfer values, the player name, goal totals, or the player's own team Elo.

04 / 30 TRANSFORMATION

From raw match logs to the analysis population

Cleaning step	Rows	Reason
Raw FBref match-log rows	2,263	Starting snapshot
Repeated header rows removed	-12	Rows such as Gl's = 'Gls'
No-minute / DNP rows removed	-47	Not valid played appearances
Exact Nations League duplicates removed	-3	Same player-date-opponent-context key
Valid appearances retained	2,201	The FAMD and per-90 population

The non-negotiable denominator

The valid set contains 1,063 scoreless appearances (48.3%). They stay in the analysis. Removing them would compare only matches in which a goal happened and would exaggerate both per-90 rates.

Input	Transformation	Output
2,201 valid appearances	One global FAMD on all valid matches	A match difficulty score
1,738 goal events	Join the match score to each goal	1,735 scored goals + 3 missing
All 181,081 valid minutes	Stay in player denominators	Raw/90 and weighted index/90

SAY IT SIMPLY A player does not vanish from the denominator merely because he did not score.

The verified baseline snapshot

Baseline means K = 1 with penalties included unless stated otherwise.

Metric	Messi	Ronaldo
Goals	847	891
Valid appearances	1,031	1,170
Valid minutes	85,131	95,950
Scoreless appearances	477	586
Penalty goals	106	169
Raw goals / 90	0.8954	0.8357
Weighted index / 90	0.1337	0.0384

Penalty-included versus penalty-excluded

Penalties	M goals	M raw/90	M weighted/90	R goals	R raw/90	R weighted/90
Included	847	0.8954	0.1337	891	0.8357	0.0384
Excluded	741	0.7834	0.0926	722	0.6772	0.0327

Overview club snapshot counts

Player	Club	Goals in this dataset
Messi	Barcelona	627
Messi	Paris Saint-Germain	32
Messi	Inter Miami	90
Ronaldo	Sporting CP	3
Ronaldo	Manchester United	127
Ronaldo	Real Madrid	421
Ronaldo	Juventus	101
Ronaldo	Al-Nassr	102

06 / 30 CALCULATION

Totals, rates, scoreless appearances, and penalties

$$\text{Raw}/90 = 90 \times \text{goals} \div \text{total valid minutes}$$

A literal scoring rate: how many goals the observed pace would produce in 90 minutes.

Symbol	Meaning	Rule
goals	Eligible goal rows for the player	Includes penalties unless the toggle excludes them
total valid minutes	Minutes across every valid appearance	Never restricted to goal-scoring matches
90	Standard match-length scale	Makes players with different minutes comparable

Worked examples

- Messi: $90 \times 847 \div 85,131 = 0.8954$ raw goals per 90.
- Ronaldo: $90 \times 891 \div 95,950 = 0.8357$ raw goals per 90.
- Without penalties, goal numerators become 741 and 722, but the minute denominators remain 85,131 and 95,950.

What each completed count means

Metric	How calculated	Interpretation
Total goals	Number of goal rows	Literal count; no context adjustment
Appearances	Valid played match rows	Includes scoreless appearances
Minutes	Sum of numeric played minutes	Denominator for every per-90 view
Scoreless	Valid appearances where GIs = 0	Prevents scorer-only selection bias
Penalties	Goal rows marked Penalty kick	Optional numerator exclusion

SAY IT SIMPLY The penalty switch removes penalty goals, not the matches or minutes in which they occurred.

07 / 30 CALCULATION

How every valid match receives an opponent Elo

Order	Source / fallback	Appearances	Purpose
1	Club Elo by opponent and date	1475	Used for matched club appearances
2	National Elo by team and date	333	Used for matched internationals
3	Competition-average Elo	186	Fallback when a direct rating is absent
4	Global 1500	207	Last fallback; no match is dropped

Coverage

- Direct club or national Elo: 1,808 of 2,201 appearances (82.1%).
- Direct or competition-average: 1,994 of 2,201 (90.6%).
- The remaining 207 appearances use 1500. All valid appearances therefore enter FAMd.

What Elo does-and does not do

- It supplies a time-specific estimate of opponent strength.
- It is only one of four FAMd inputs; it is not itself the final weight.
- The player's own team Elo is deliberately excluded, avoiding a penalty for playing in a stronger side or era.
- Fallback values preserve coverage but are less specific than direct historical ratings.

SAY IT SIMPLY First use the opponent's historical rating; if it is missing, use the best documented fallback and keep the match.

One global FAMD turns mixed match context into a common index

D_i = standardized FAMD Dim 1 score

The same global model is fitted once to all 2,201 valid appearances for both players.

Input	Kind	Dim 1 contribution	Role
Opponent_Elo	continuous	0.8%	Opponent strength
Venue	categorical	48.6%	Home / Away / Neutral
Competition_Stage	categorical	2.3%	Group / knockout / final / qualifying / other
Is_Away	categorical	48.3%	Explicit away indicator

Orientation and standardization

- FAMD is used because the inputs mix one continuous variable with categorical variables; PCA alone is not appropriate for that mixture.
- Dim 1 is flipped when necessary so its correlation with opponent Elo is positive: higher is intended to mean harder.
- Dim 1 is then centred and scaled across valid appearances. Zero is the centre of this index, not 'no difficulty'.
- The first dimension explains 25.3% of total variation. Most variation is therefore outside this one-number summary.

MAJOR CAUTION Venue (48.6%) and Is_Away (48.3%) overlap conceptually. Together they dominate Dim 1, while Opponent Elo contributes only 0.8%. The current index is largely an away/home axis-not a pure opponent-strength measure.

From a match index to a selected-K weighted rate

$$WG_i = D_i$$

Base weighted goal at K = 1. Each scored goal inherits its match's difficulty index.

$$WG_i(K) = \text{sign}(D_i) \times |D_i|^K$$

Signed power preserves the sign and remains finite for negative scores at fractional K.

$$\text{Weighted}/90 = 90 \times \Sigma WG_i(K) \div \Sigma \text{ minutes}$$

Goal-index contributions form the numerator; all valid appearance minutes form the denominator.

$$\text{Gap}(K) = \text{Messi Weighted}/90 - \text{Ronaldo Weighted}/90$$

A positive gap means Messi's index rate is higher at that tested K; a negative gap would mean Ronaldo's is higher.

Symbol	Meaning
i	A goal with an available match difficulty score
D _i	Oriented, standardized FAMD Dim 1 score
K	Sensitivity exponent from 0.5 to 3.0
sign(D _i)	+1 for positive scores, -1 for negative scores
Σ minutes	All valid player minutes, including scoreless appearances

Worked signed-power example

If D_i = -0.64 and K = 0.5, then WG_i(K) = -1 × |-0.64|^{0.5} = -0.80. Plain (-0.64)^{0.5} would be invalid in ordinary real-number calculation; signed power is why the dashboard remains finite.

K sensitivity with penalties included

K	Messi weighted/90	Ronaldo weighted/90	Messi - Ronaldo gap
0.5	0.1768	0.0905	+0.0863
1.0	0.1337	0.0384	+0.0953
1.5	0.0940	-0.0104	+0.1044
3.0	-0.0009	-0.1389	+0.1380

How to read the pattern

- K below 1 compresses large absolute scores and lifts smaller absolute values relative to them.
- K = 1 preserves the base difficulty index.
- K above 1 amplifies large absolute scores and shrinks values whose absolute size is below 1.
- At K = 3, both rates can cross below zero: Messi -0.0009 and Ronaldo -0.1389. These are valid centred-index outputs.

VERIFIED STABILITY RESULT No lead change occurs at any of the 26 tested values from K = 0.5 to 3.0. The positive gap ranges from 0.0863 to 0.1380 index units per 90.

What the test does not establish

It does not attach sampling uncertainty, prove causation, or show that the FAMD inputs are the only valid definition of difficulty. It asks a narrower question: does the ordering change when the exponent changes on this fixed dataset and index?

K sensitivity with penalties excluded

K	Messi weighted/90	Ronaldo weighted/90	Messi - Ronaldo gap
0.5	0.1312	0.0739	+0.0573
1.0	0.0926	0.0327	+0.0600
1.5	0.0569	-0.0059	+0.0627
3.0	-0.0306	-0.1066	+0.0760

The denominator contract

- Eligible goal numerators become 741 for Messi and 722 for Ronaldo.
- Minute denominators remain 85,131 and 95,950; the toggle never removes appearances.
- At K = 3 the valid rates are -0.0306 and -0.1066, not errors.

VERIFIED STABILITY RESULT No lead change occurs at any of the same 26 K values without penalties. The positive gap ranges from 0.0573 to 0.0760 index units per 90.

Lead stability calculation

For each penalty setting, the dashboard computes Gap(K) at K = 0.5, 0.6, ..., 3.0. It checks whether adjacent gaps touch zero or change sign. Here all 26 gaps are positive in both settings, so the label is “No lead change across K=0.5-3.0.”

SAY IT SIMPLY The ordering survives this exponent stress test, but that is not the same as proving a population-level difference.

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Base-score density, fixed deciles, counts, and shares

Penalty setting	Messi median	Ronaldo median	Messi D10	Ronaldo D10
Included	0.808	0.750	99 (11.7%)	74 (8.3%)
Excluded	0.804	0.750	81 (11.0%)	63 (8.7%)

Normalized density

- A smooth curve estimates the distribution of K=1 difficulty scores for each player's eligible goals.
- Each player's curve is normalized to area 1. Curve height is relative concentration, not a goal count.
- The density stays on the base score; moving K does not change it. The penalty toggle changes which goals contribute.

Fixed global deciles

- The 1,735 goals with a score are ranked together before penalty filtering.
- Ranks are split into D1 (easiest) through D10 (hardest). Ties use a stable first-occurrence rule.
- The D10 base-score range is 0.993 to 1.943 and does not move when penalties are excluded.
- Count = eligible goals in a decile. Player share = that count ÷ all eligible scored goals for the same player.

Player share in D10 = D10 eligible goals ÷ all eligible scored goals

Included example: Messi = $99 \div 845 = 11.7\%$; Ronaldo = $74 \div 890 = 8.3\%$.

SAY IT SIMPLY Density compares distribution shape; decile bars compare counts. They answer different questions.

Caveats deserve the same prominence as the results

Caveat	What it means
Three missing difficulty scores	Raw rates include all 1,738 goals. Weighted views use 1,735; 2 Messi goals and 1 Ronaldo goal lack match context.
xG coverage	1,255 goals lack xG. xG is attached but does not enter the current weight.
One dimension is partial	Dim 1 explains 25.3% of total FAMD variation; 74.7% is not represented by this index.
Away/home dominance	Venue contributes 48.6% and Is_Away 48.3%; their overlap makes Dim 1 largely an away/home axis.
Opponent Elo is small	Opponent Elo contributes 0.8% to Dim 1, so the score should not be described as mainly opponent strength.
Negative values are valid	They are centred index values, not goals being taken away from the historical record.
K stability is narrow	It is a sensitivity check on one transformation, not a causal estimate or winner declaration.
Bootstrap scope is conditional	The interval describes match-resampling uncertainty under the clicked K, penalty, competition, venue, dataset, and weighting definition.
Snapshot limits	The data are a collected snapshot and may differ from live official career totals or alternative providers.

RESPONSIBLE CONCLUSION Wave 4 quantifies match-resampling uncertainty under the frozen dashboard assumptions. It does not settle every football definition of difficulty, remove model limitations, or declare a final winner.

Best neutral wording

“At the default settings, the observed Messi-minus-Ronaldo weighted-index gap is 0.0953 per 90. The seeded 95% match-bootstrap interval is [-0.0120, 0.1999], and 96.0% of resampled gaps are above zero. These values are conditional on the dataset and index-not a final winner declaration.”

How the career trajectories are aligned and calculated

Default view: K = 1, penalties included, all competitions, all venues, age axis.

Cumulative at t = Σ selected-K contributions through appearance t

The curve adds each eligible goal's signed-power contribution in filtered appearance order. Scoreless appearances contribute zero and stay on the timeline.

Rolling-30/90 = 90 × Σ contributions in latest 30 appearances ÷ Σ minutes in those 30

The first 29 filtered appearances have no rolling value because a full 30-appearance window does not yet exist.

Default endpoint	Messi	Ronaldo
Cumulative weighted index	126.4402	40.9562
Latest rolling-30 index / 90	-0.0474	0.0239

Age versus calendar date

- Age aligns the careers by life stage rather than by the year on the calendar.
- Age is derived from fixed display metadata: Messi 1987-06-24; Ronaldo 1985-02-05.
- The age-30 guide appears only on the age axis. Calendar date preserves the actual historical sequence.
- Changing competition or venue restarts the cumulative line, appearance count, and rolling window inside that selected scope.

INTERPRET CAREFULLY Cumulative values answer how contributions accumulated in the selected scope. Rolling values answer what the latest 30 filtered appearances looked like. They are different questions, and neither is a confidence interval.

Opponent Elo scatter and descriptive LOESS

$$\text{Marker } y = \text{sign}(\text{Difficulty Score}) \times |\text{Difficulty Score}|^K$$

Each marker is one eligible goal. The x-axis is the actual unbinned opponent Elo; no jitter is applied.

Player	Eligible goal markers	Distinct Elo values	Default cumulative contribution
Messi	845	449	126.4402
Ronaldo	890	465	40.9562

What the chart can show

- Horizontal position shows the opponent rating attached to that goal's match.
- Vertical position shows the selected-K index contribution, which may be negative because the score is centred.
- The tooltip retains player, date, opponent, competition, venue, Elo, base score, selected-K contribution, and penalty status.
- A player-specific LOESS curve summarizes the local shape using the actual marker values.

What LOESS does not show

- It is a descriptive smoother, not a causal model, significance test, or prediction interval.
- It has no confidence band in the Head-to-Head view.
- It is suppressed when fewer than 10 goals or 5 distinct Elo values remain, because sparse smoothers are misleading.

SAY IT SIMPLY The dots are the observed eligible goals; the curve is only a visual summary of their pattern.

Penalty dependency, filters, and sparse-scope rules

Player	Penalty goals	Open-play goals	Penalty share
Messi	106	741	12.5%
Ronaldo	169	722	19.0%

- The composition chart always discloses both goal types. When penalties are excluded, the penalty segment is muted and labeled excluded rather than hidden.
- Penalty exclusion changes goal contributions only; it never removes valid appearances or minutes.
- Weighted coverage is 1,735 of 1,738 goals (99.8%). Missing-score goals remain in raw composition counts.
- The shared filters offer 31 competition labels plus All competitions, and Home, Away, Neutral, plus All venues.

Verified filter examples

Scope	Messi	Ronaldo	Why it matters
Champions Lg + Away	81 apps / 49 goals	92 apps / 63 goals	Both players remain
World Cup + Neutral	29 apps / 21 goals	25 apps / 11 goals	Neutral international scope
MLS + All	76 apps / 69 goals	0 apps	Ronaldo is N/A
FA Cup + All	0 apps	1 app / 0 goals	Messi N/A; Ronaldo 0.0000
MLS + Neutral	0 apps	0 apps	Explanatory empty state

N/A VERSUS ZERO N/A means a player has no valid appearances in the selected scope. 0.0000 means appearances exist but no eligible weighted contribution remains. Keeping those states separate prevents a missing comparison from looking like a real zero.

WAVE 4 HANDOFF Head-to-Head remains descriptive and updates immediately. Inference changes only when Update analysis is clicked, then displays the frozen settings and warns if the live controls change.

How the match-level bootstrap measures uncertainty

Default frozen scope: K = 1, penalties included, all competitions, all venues.

$$\text{Rate}^* = 90 \times \Sigma \text{resampled match contributions} \div \Sigma \text{resampled match minutes}$$

Each player is resampled independently with replacement, using exactly that player's selected-scope appearance count.

$$\text{Gap}^* = \text{Messi Rate}^* - \text{Ronaldo Rate}^*$$

Repeat 10,000 times. The 2.5th and 97.5th percentiles form the displayed 95% interval.

$$P(\text{Gap}^* > 0) = \text{count}(\text{Gap}^* > 0) \div 10,000$$

A directional share of the bootstrap distribution, not a p-value or binary decision.

Default result	Verified value	How to read it
Observed ratio-of-sums gap / 90	+0.0953	Messi - Ronaldo under frozen settings
95% percentile interval	[-0.0120, +0.1999]	Middle 95% of seeded resampled gaps
P(gap > 0)	0.960	Share of resampled gaps above zero
Match-level Cohen's d	+0.090	Pooled-SD appearance-rate difference
Replicates / seed	10,000 / 20260812	Deterministic and fully finite

INTERVAL INTERPRETATION The default percentile interval crosses zero. Report that numerical fact and the full distribution; do not convert it into a winner, significant/not-significant badge, or p-value.

Career-era subgroup results

Every row inherits the clicked global competition and venue scope.

Era	Messi apps / rate	Ronaldo apps / rate	Gap	95% interval	d
Rise (2002-08)	102 / 0.0963	274 / 0.0677	+0.0286	[-0.1915, +0.2416]	+0.142
Peak (2009-14)	288 / 0.0005	300 / 0.0128	-0.0124	[-0.2397, +0.2181]	-0.024
Prime (2015-18)	248 / 0.1936	224 / 0.1344	+0.0592	[-0.1890, +0.3123]	+0.042
Transition (2019-22)	208 / 0.2246	207 / 0.0598	+0.1649	[-0.0653, +0.3909]	+0.162
Late (2023-26)	185 / 0.1701	165 / -0.1192	+0.2893	[+0.0307, +0.5535]	+0.267

- Rates are descriptive whenever appearances exist.
- Intervals and Cohen's d require at least two appearances for both players.
- A row is marked sparse in the dashboard when either player has fewer than 30 appearances.
- Era comparisons are exploratory; they are not adjusted for repeated subgroup inspection.

SUBGROUP CAUTION Differences across eras can reflect career timing, teams, competitions, and the index definition. The table quantifies conditional patterns; it does not isolate a causal era effect.

19 / 30 INFERENCE

Competition-family results and Cohen's d

Family	Messi apps / rate	Ronaldo apps / rate	Gap	95% interval	d	Count
League	654 / 0.0013	758 / -0.0106	+0.0120	[-0.1264, +0.1500]	+0.043	Adequate count
Continental	192 / 0.1744	198 / 0.0985	+0.0759	[-0.1752, +0.3291]	+0.024	Adequate count
Domestic cup	36 / 0.6210	14 / -0.1330	+0.7540	[+0.2088, +1.3315]	+0.756	Sparse
International	137 / 0.5416	196 / 0.1662	+0.3754	[+0.1174, +0.6423]	+0.330	Adequate count
Other	12 / 0.3401	4 / 0.4289	-0.0888	[-2.0558, +0.5513]	-0.686	Sparse

$$d = (\text{mean Messi appearance rate} - \text{mean Ronaldo appearance rate}) \div \text{pooled SD}$$

This conventional match-level effect size uses appearance-level weighted rates. It is not the primary ratio-of-sums gap and does not use the bootstrap means.

- Domestic cup and Other are sparse at the default scope because at least one player has fewer than 30 appearances.
- If an exact competition is selected, only its inherited competition family may remain.
- Scoreless appearances stay in both family denominators; unmapped scoreless-only competition labels fall into Other.
- N/A is distinct from zero: no appearances suppress a comparison, while real appearances with no eligible contribution can produce 0.0000.

20 / 30 FIELD APPENDIX A

Master goal table: fields 1-15 of 29

File: data/processed/goals_master_final.csv • one row per goal

Group	Field	Type	What it means	Why we need it
Identity and match	Player	text	Player name.	Separates the two careers.
Identity and match	Rk	text	FBref row number.	Supports source auditing.
Identity and match	Date	date	Match date.	Joins goals to match context and Elo.
Identity and match	Comp	text	Competition name.	Identifies football context.
Identity and match	Round	text	Round or stage label.	Builds Competition_Stage.
Identity and match	Venue	text	Home, Away, or Neutral.	One FAMd input.
Identity and match	Squad	text	Source team label.	Preserves the collected value.
Identity and match	Opponent	text	Source opponent label.	Preserves the collected value.
Identity and match	Start	text	Whether the player started.	Describes the appearance.
Goal event	Minute	text	Goal minute, including stoppage time.	Orders multiple goals in a match.
Goal event	Score	text	Score immediately after the goal.	Describes match state.
Goal event	Goalkeeper	text	Goalkeeper beaten.	Event-level context.
Goal event	Assist	text	Assisting player when recorded.	Event-level context.
Goal event	Notes	text	Goal note, such as Penalty kick.	Creates the penalty flag.
Clean joins and match context	Opp_clean	text	Opponent without country prefix.	Makes opponent joins reliable.

APPENDIX READING RULE A field can be present for traceability without entering the current formula. The final column says whether its purpose is analysis, joining, display, or auditing.

Master goal table: fields 16-29 of 29

Group	Field	Type	What it means	Why we need it
Clean joins and match context	Squad_clean	text	Team without country prefix.	Supports club and national labels.
Clean joins and match context	Venue_match	text	Venue copied from the match log.	Cross-checks the goal-log venue.
Clean joins and match context	Result	text	Full-time match result.	Adds match outcome context.
Clean joins and match context	Minutes	number	Minutes in the appearance.	Supplies the per-90 denominator.
Clean joins and match context	Gls	number	Goals in that match, including zero.	Retains scoreless appearances in match data.
Difficulty, xG, and identifiers	Opponent_Elo	number	Opponent rating or fallback value.	Quantitative FAMD input.
Difficulty, xG, and identifiers	Elo_Source	text	club, national, league_avg, or global_1500.	Discloses how the rating was obtained.
Difficulty, xG, and identifiers	Competition_Stage	text	Group, Knockout, Final, Qualifying, or Other.	Categorical FAMD input.
Difficulty, xG, and identifiers	Is_Away	logical	TRUE for an away match.	Binary FAMD input.
Difficulty, xG, and identifiers	Difficulty_Score	number	Oriented, standardized global FAMD Dim 1.	Base difficulty index for weighting.
Difficulty, xG, and identifiers	xg	number	Understat expected-goal value, when matched.	Attached context; not used in the weight.
Difficulty, xG, and identifiers	xg_source	text	understat or missing.	Makes xG coverage explicit.
Difficulty, xG, and identifiers	goal_id	text	Unique goal identifier.	Tracks one row without ambiguity.
Difficulty, xG, and identifiers	Weighted_Goal	number	Base value equal to Difficulty_Score.	Stores the K=1 weighted index contribution.

Three distinctions that prevent mistakes

- Venue is the goal-log value; Venue_match is the joined match-log copy used as a consistency check.
- Opponent_Elo is the chosen value; Elo_Source records whether it was direct or a fallback.
- Weighted_Goal is the base K=1 index value. The selected-K value is computed interactively and is not written back to the master CSV.

COVERAGE NOTE The master table has 29 columns. The dashboard adds six fields in memory, shown on the next page; they do not alter the source CSV.

Six dashboard-derived goal fields and valid-match fields 1-10

Group	Field	Type	What it means	Why we need it
Dashboard-derived goal fields	Season	integer	Season-ending year derived from Date.	Supports time filters.
Dashboard-derived goal fields	Era	ordered factor	Rise, Peak, Prime, Transition, or Late.	Creates consistent career periods.
Dashboard-derived goal fields	Penalty	logical	TRUE when Notes says Penalty kick.	Runs included/excluded views.
Dashboard-derived goal fields	Comp_Group	text	League, Continental, Domestic cup, International, or Other.	Simplifies competition filters.
Dashboard-derived goal fields	is_intl	logical	TRUE for national-team appearances.	Separates club and international context.
Dashboard-derived goal fields	Club_Goal	logical	TRUE when is_intl is FALSE.	Builds the Overview club snapshot.

Valid-match table: fields 1-10 of 42

Group	Field	Type	What it means	Why we need it
Identity and schedule	Player	text	Player name.	Separates careers.
Identity and schedule	Date	date	Match date.	Aligns matches, goals, and Elo.
Identity and schedule	Day	text	Day of week.	Source context and auditing.
Identity and schedule	Comp	text	Competition.	Groups football contexts.
Identity and schedule	Round	text	Round or matchweek.	Builds stage categories.
Identity and schedule	Venue	text	Home, Away, or Neutral.	FAMD input.
Identity and schedule	Result	text	Full-time result.	Describes the appearance.
Identity and schedule	Squad	text	Source team label.	Identifies the player's side.
Identity and schedule	Opponent	text	Source opponent label.	Identifies the opposing side.
Identity and schedule	Start	text	Starter indicator.	Describes playing role.

Valid-match table: fields 11-26 of 42

This is the analysis population; zero-goal appearances remain.

Group	Field	Type	What it means	Why we need it
Identity and schedule	Pos	text	Recorded position.	Appearance context; not in FAMd.
Identity and schedule	Min	text	Original minute value.	Preserves the raw source field.
Attacking match statistics	Gls	number	Goals in the appearance; zero retained.	Counts scoring and scoreless matches.
Attacking match statistics	Ast	number	Assists.	Context only; not in the current weight.
Attacking match statistics	PK	number	Penalties scored.	Source check for penalty context.
Attacking match statistics	PKatt	number	Penalties attempted.	Source check for penalty context.
Attacking match statistics	Sh	number	Shots.	Context only.
Attacking match statistics	SoT	number	Shots on target.	Context only.
Discipline and actions	CrY	number	Yellow cards.	Source context only.
Discipline and actions	CrR	number	Red cards.	Source context only.
Discipline and actions	Fls	number	Fouls committed.	Source context only.
Discipline and actions	Fld	number	Fouls drawn.	Source context only.
Discipline and actions	Off	number	Offsides.	Source context only.
Discipline and actions	Crs	number	Crosses.	Source context only.
Discipline and actions	TklW	number	Tackles won.	Source context only.
Discipline and actions	Int	number	Interceptions.	Source context only.

WHY MANY FIELDS ARE NOT INPUTS The valid-match record keeps source context for auditing. Only Opponent_Elo, Venue, Competition_Stage, and Is_Away enter FAMd.

Valid-match table: fields 27-42 of 42

Group	Field	Type	What it means	Why we need it
Discipline and actions	OG	number	Own goals.	Source context only.
Discipline and actions	PKwon	number	Penalties won.	Source context only.
Discipline and actions	PKcon	number	Penalties conceded.	Source context only.
Discipline and actions	Match Report	text	FBref match-report link/value.	Traceability to the source.
Cleaned and analytical fields	Season_End	integer	Season label from collection.	Preserves source season grouping.
Cleaned and analytical fields	Squad_clean	text	Clean team name.	Reliable joins and display.
Cleaned and analytical fields	Opp_clean	text	Clean opponent name.	Reliable joins to Elo.
Cleaned and analytical fields	Minutes	number	Numeric minutes played.	Complete per-90 denominator.
Cleaned and analytical fields	is_intl	logical	National-team match indicator.	Chooses national versus club Elo.
Cleaned and analytical fields	Competition_Stage	text	Derived stage category.	FAMD input.
Cleaned and analytical fields	Is_Away	logical	Away-match indicator.	FAMD input.
Cleaned and analytical fields	Opponent_Elo	number	Opponent rating after fallback.	FAMD input.
Cleaned and analytical fields	Elo_Source	text	Rating source or fallback.	Audits rating coverage.
Cleaned and analytical fields	Difficulty_Score	number	Standardized global FAMD Dim 1.	Joined to goals for weighting.
Cleaned and analytical fields	Season	integer	Season-ending year from Date.	Dashboard time grouping.
Cleaned and analytical fields	Era	ordered factor	Five-part career era.	Dashboard era grouping.

DENOMINATOR REMINDER Minutes is numeric and valid for all 2,201 retained rows. Gl's may be zero; that is expected and required.

Plain-English glossary

Term	Meaning in this project
Appearance	A match in which the player recorded positive playing time.
Bootstrap	Resampling whole appearances with replacement to describe uncertainty in the weighted-rate gap.
Cohen's d	A standardized difference between the two sets of appearance-level weighted rates.
Directional probability	The share of bootstrap gaps above zero; not a p-value.
LOESS	A descriptive local smoother used to summarize the Elo scatter; not an inferential model.
Decile	One of ten ranked groups; D1 is easiest and D10 hardest here.
Denominator	The quantity divided by—in this guide, all valid player minutes.
Density	A smoothed, normalized picture of where scores are concentrated.
Dim 1	The first FAMD axis: the single summary used as difficulty index.
Elo	A rating that changes over time and estimates team strength.
FAMD	Factor Analysis of Mixed Data, for continuous and categorical inputs together.
Index	A constructed scale for comparison; not a literal count or physical unit.
K	The exponent controlling sensitivity to the size of difficulty scores.
Normalized	Rescaled so comparisons concern shape rather than sample size.
Per 90	A rate scaled to 90 minutes using all valid minutes.
Sensitivity check	A test of whether a conclusion changes when an assumption changes.
Signed power	$\text{sign}(x) \times x ^k$, which preserves negative-score direction.
Standardized	Centred at approximately zero and scaled to standard-deviation units.
Weighted goal	Here, a goal's index contribution—not a literal fractional goal.
xG	Expected goals: a shot-quality estimate; attached but not used in this weight.

LANGUAGE CHOICE Say “weighted index units per 90,” especially when values are negative. That phrasing avoids turning an abstract index into imaginary goals.

How to explain this to a football fan

Five speaking notes

- Start with playing time: both raw and weighted rates use every valid minute, including scoreless matches.
- Separate literal facts from index results: goals and raw/90 are counts/rates; weighted/90 is an abstract context index.
- Explain the source of difficulty: one global FAMD combines Elo, venue, stage, and away status for both players.
- Explain Head-to-Head scope: competition and venue filter both goal contributions and all valid appearance minutes; penalty exclusion changes only goal contributions.
- End with the limits: the bootstrap quantifies conditional match-resampling uncertainty but does not repair the index's conceptual limits or declare a winner.

Common misunderstandings and answers

Misunderstanding	Answer
"A negative value means negative goals."	No. It is a centred index contribution; historical goal counts never change.
"The model says Elo barely matters, so opponent strength is irrelevant."	No. It says Elo contributes little to this particular Dim 1; other dimensions and models may represent it differently.
"Penalty exclusion should remove those match minutes."	No. The toggle changes goal numerators only; played minutes remain valid exposure.
"No lead change means statistical proof."	No. It is an exponent sensitivity result; Wave 4 separately reports the match-bootstrap distribution.
"The 96% directional probability is a p-value."	No. It is the fraction of resampled gaps above zero under the frozen assumptions.
"The dashboard has xG, so xG creates the weight."	No. xG is attached for 483 goals but is not an FAMD input or weighting term.
"The LOESS curve proves an Elo effect."	No. It is an unbinned descriptive smoother without a confidence band or causal interpretation.
"N/A and 0.0000 mean the same thing."	No. N/A means no appearances exist; zero means appearances exist but no eligible contribution remains.

WAVE 5 AND 6 BRIDGE The next four pages document the completed Methodology, Summary, Raw Data, and interface-polish work. They preserve the same fixed formulas, all-valid-minute denominators, and source-data contract.

USE THE EVIDENCE • STATE THE LIMITS • DO NOT DECLARE A FINAL WINNER

The complete analysis map, from source rows to uncertainty

Step	Dashboard evidence	Statistical contract
1. Collect and clean	1,738 goals; 2,201 valid appearances	Keep scoreless appearances and all positive-minute exposure
2. Build context	One global FAMD for both players	Inputs only: Opponent Elo, Venue, Competition Stage, Is Away
3. Weight goals	Stored Dim 1 score; selected signed-power K	Missing difficulty contributes zero; negative direction is preserved
4. Scale per 90	181,081 total valid minutes	Ratio of sums, never goal-scoring matches only
5. Quantify uncertainty	10,000 match resamples; seed 20260812	Independent within-player resampling; report the full gap distribution

What creates the context index

FAMD input	Dim 1 contribution
Opponent Elo	0.8013%
Venue	48.6372%
Competition Stage	2.2663%
Is Away	48.2952%

$$WG_i(K) = \text{sign}(D_i) \times |D_i|^K$$

The same signed-power rule appears in the sidebar and Methodology tab. K changes sensitivity; it does not re-run FAMD.

INTERPRETATION LIMIT Dim 1 explains 25.2968% of the mixed-data variation and is a centred analytical index. Negative values are not negative goals, and a larger contribution share is not causal importance.

28 / 30 WAVE 5 SUMMARY

The fixed career reference and venue comparison

Scope	Messi apps / min	Ronaldo apps / min	Weighted / 90 (M / R / gap)	Raw / 90 (M / R / gap)
Overall	1,031 / 85,131	1,170 / 95,950	0.1337 / 0.0384 / +0.0953	0.8954 / 0.8357 / +0.0597
Home	495 / 40,525	572 / 47,198	0.9184 / 0.7791 / +0.1393	1.0638 / 0.9325 / +0.1313
Away	461 / 38,180	538 / 43,531	-0.8123 / -0.8450 / +0.0327	0.7472 / 0.7588 / -0.0115
Neutral	75 / 6,426	60 / 5,221	0.8056 / 0.7080 / +0.0976	0.7143 / 0.6033 / +0.1110

- The fixed reference contains 14 ordered rows: overall, five career eras, five competition families, and three venues.
- The fixed table never changes. The live table updates immediately with K, penalty, competition, and venue controls; it does not wait for Update analysis.
- Weighted and raw columns remain side by side. N/A means missing player exposure; a real zero is printed as 0.0000.
- Apps and minutes reconcile to the same valid-match denominators used everywhere else.

READING RULE Every gap is Messi minus Ronaldo. These Summary rows are descriptive; uncertainty remains on the button-frozen Inference tab.

Exact source transparency and the downloadable audit trail

Raw-data contract	Verified value
Rows × columns	1,738 × 29
File size	431,529 bytes
Source/app MD5	c43c3f995b1f301b4328c846eab2cf27
Download behavior	Byte-for-byte copy of goals_master_final.csv
Dashboard controls	No effect on rows, search, sorting, filters, or download

How to audit the table

- Use the global search to find a value across all 29 fields.
- Use the per-column search boxes to narrow a specific source field.
- Sort any header and paginate through all 1,738 goal-event rows.
- Scroll horizontally inside the table on narrow screens; the page itself remains contained.
- Download the same preserved CSV used by the tab, with no current sidebar filters applied.

NEXT PAGE Wave 5 completed the analytical content. The final page explains Wave 6 interface polish and separates usability improvements from statistical changes.

How the finished interface supports careful reading

Area	What changed	Why it helps
Loading feedback	Native Shiny busy indicators plus polite updating/updated announcements	Shows when outputs are recalculating without adding a dependency
Plot caching	All eight Plotly outputs use dependency-aware caches	Avoids unnecessary redraw work; cached results do not change the statistic
Sidebar behavior	Open on desktop and initially collapsed on mobile	Keeps controls ready on wide screens and preserves space on narrow screens
Keyboard access	Visible skip link, focus target, and focus-visible styling	Lets keyboard users move directly to dashboard content
Responsive layout	Contained tables, formulas, and Plotly controls at 1919, 768, 390, and 320 px	Prevents page-level horizontal overflow
Motion preference	Reduced-motion rules suppress animated loading treatments	Respects the reader's operating-system preference

What Wave 6 did not change

- No source row, analysis-bundle field, global FAMD score, formula, denominator, or seeded bootstrap result changed.
- The signed-power rule remains $\text{sign}(\text{Difficulty_Score}) \times |\text{Difficulty_Score}|^k$.
- All 2,201 valid appearances and 181,081 minutes remain the exposure base, including 1,063 scoreless appearances.
- Inference still changes only when Update analysis is clicked and still uses 10,000 match-level resamples.
- The app still requires only shiny, bslib, data.table, htmltools, plotly, and DT at runtime.

STOPPING POINT Wave 6 is complete and verified. The analytical dashboard and its usability polish are ready for local review. Containerization and deployment remain in later review-gated waves and have not been started.

STOP AT THE WAVE 6 REVIEW GATE